Smart Lab - Round Report and Remaining Work

17 October 2026 (night) · Portal **http://192.168.1.200** · nothing committed or pushed · follows `SmartLab_Continuation_2026-10-17.pdf`

1. Done in this round

Item	What changed	Measured / tested	Status
Security → Live: both cameras at once	When a lab has more than one camera, Live shows all of them side by side at full width, each labelled with its device ID and status. Click one to show its details. A checkbox switches back to a single camera.	Live portal: both LAB_01 feeds shown; camera 2 live, camera 1 "no picture" (at the university).	**VERIFIED**
Lab Equipment → Robots & PCs lag	Cause: a once-per-second clock re-rendered both 3D robot panels and their movement-replay lists (up to 1,000 rows). Panels now re-render only when the robot's data changes. The replay list is memoised. The PC camera poll is slowed from every 3 s to every 5 s.	Main-thread blocking per 10 s: **10.5 s → 2.8 s**. Long tasks: 74 → 13. Tab opens in 3.8 s.	**VERIFIED**
Home / STOP on both robots	The master-control panel now also sits under each twin in Lab Equipment (before, it was only on each robot's own page). UR5e: STOP ROBOT, Home, exact Home pose; BLOCKED until live joint data; DRY RUN. FANUC: Home/Stop disabled, with the reason (no verified remote interface).	Live: 1 STOP button, 3 Home buttons, FANUC reason shown.	**VERIFIED (UI); robot motion still DRY RUN**
IoT Dashboard robot-fleet bug	The small twin thumbnails drew the full view bar (Front/Back/...) and status text over the card. Both are hidden on thumbnails; the full twin keeps them.	Live: 0 view buttons in the fleet card; screenshot clean.	**VERIFIED**
Repeated camera warnings	Checked against the data: not duplicates. Each incident is a different PC box, and repeats already fold into one per box. New rule: 3 or more boxes of one camera with the same warning within 5 s → one "camera-wide change" incident listing the boxes. Every event is still stored.	New test passes; security tests 14/14.	**VERIFIED**
Priority incidents review	Evidence for the 6 "Forced entry" and 2 "Person with no door entry" incidents, written up in `docs/INCIDENT_REVIEW_2026-10-17.md`. Most fall seconds after granted entries or controller restarts during bring-up. #84 (3 Oct 03:22) has nothing around it. Nothing was changed in the portal.	Read-only database queries.	**PARTIAL: needs your decisions**
Docker restart	Docker Desktop updated itself mid-session (to 29.8.2) and stopped all containers. The stack was started again as before (database, backend, frontend, n8n, backup).	Health 200; all containers up.	**VERIFIED**

2. Tests and evidence

- **Backend:** full suite **495 passed** (includes the new camera-wide test).
- **Frontend:** TypeScript check clean, deployed, live pages checked.
- **Evidence run** `docs/reports/evidence_20261017_194939.md`: 33 pass, 14 fail, 7 not available, 2 simulation. Twelve failures are hardware that's off or away. The two new ones:
  - The admin-laptop PC agent is offline. Expected: the test agent ended with the earlier session.

- "What is the UR5e home position" took **3.8 s** on the first, uncached call; the limit is 3 s. The cause is the search embedding model, which Ollama unloads after 15 idle minutes, so the first question after a quiet spell reloads it. Repeat calls take 0.03-0.04 s and give the correct angles, citing ROBOT_CONTROL.md.

3. Everything remaining

Item	Status	Exact next action
AI first-question delay after idle	**READY TO APPLY**	In `tools/knowledge/kb.py` change the embedding `keep_alive` from "15m" to "2h", the same as the answer model. Then restart the knowledge service, which needs your OK.
PC agent on a student PC	**BLOCKED (IT)**	IT approval → copy `tools/pc_agent`, fill `agent.env`, run `start_pc_agent.bat` → test View screen, monitor switch, wall, message, open app, share to that PC, Stop. The admin laptop is already verified.
Remote keyboard/mouse	**BLOCKED (IT)**	Not built. Use Windows RDP shadowing with the user's consent, or an IT-approved remote-support tool.
UR5e live Home / STOP	**PHYSICAL VALIDATION**	Cable + IP .102 → live joints in the twin → pendant Home shows "Home reached" → clearance check → Remote mode → your written sign-off → `ROBOT_CONTROL=live`, backend restart only. STOP is software only, never an emergency stop.
FANUC remote interface	**BLOCKED (lab)**	On the pendant: MENU → STATUS → Version ID and ORDER FI. Photograph the options list (R648 / R641 / KAREL / SNPX / PNS-UOP) and write the HOME TP program. Then the adapter can be designed.
Camera 1 (MC210)	**PHYSICAL VALIDATION**	At the lab: ping .155, check the picture in Security → Live.
Camera 2 (MC200)	**PHYSICAL VALIDATION**	Mount it diagonally opposite camera 1 → `floorcal.py` → walk / crossing / occlusion / lights / fast-walk tests; note the stream delay.
ESP32 boards (door, environment, 2 bins)	**PHYSICAL VALIDATION**	Power them on the Smart_Lab Wi-Fi; check ONLINE on System health. Their "offline" incidents clear by themselves when they report again.
Flame detector (17% history)	**PHYSICAL VALIDATION**	Bench test: cover the sensor; flame at 50 cm; phone torch at 30 cm (if the torch triggers it, turn the potentiometer down). Record the result in the incident. No threshold change before the test.
Inventory discrepancy	**BLOCKED (staff count)**	One physical count per bin (Lab Equipment → Consumables → Count). Unknown stays unknown until then.
Incident triage	**NEEDS YOU**	In Incidents, with "group similar" on: decide the 6 forced entries (#84 first) and the 2 person-no-entry (view their snapshots in the portal only); close each with a real reason.
Commit	**NEEDS YOU**	All work is uncommitted on `offline-router-door-photos`, with backups in `backups/`. Say "commit" when you want it (nothing is pushed).

4. Files changed in this round

- `backend/app/api/routes/security.py`, `backend/tests/test_security.py`
- `frontend/src/pages/Security.tsx`, `components/robot/RobotTwin.tsx`, `components/robot/RobotTwinPanel.tsx`, `components/robot/Replay.tsx`
- `docs/INCIDENT_REVIEW_2026-10-17.md`, `docs/LAB_READINESS_2026-10-18.md` (update section), `docs/PROGRESS_CHECKPOINT.md`